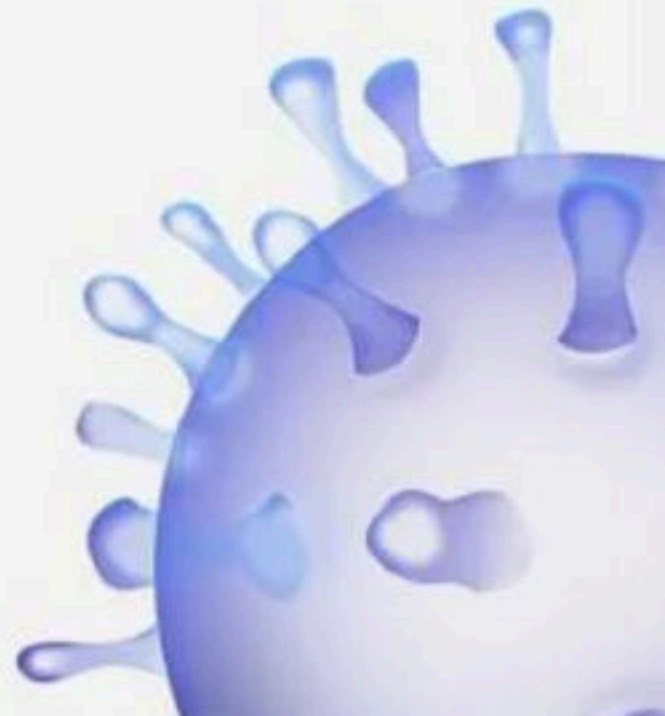


Locomotion and Movement - 100MCQs

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Locomotion and Movement



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1. Which of the following options shows correct order of some stages of muscle contraction from the beginning to the end of the process?

- (1) Stimuli, Neurotransmitter secretion, cross bridges formation, Excitation of T-system, Sliding of actin filaments.
- (2) Stimuli, Excitation of T-system, Neurotransmitter secretion, cross bridges formation, Sliding of actin filaments, 'H'- band diminishes
- (3) Stimuli, neurotransmitter secretion, Release of Ca^{2+} , cross bridges formation, Excitation of T-system Sliding of actin filaments.
- (4) Stimuli, Neurotransmitter secretion, Excitation of Tsystem release of Ca^{2+} cross, bridges formation, Sliding of actin filaments, 'H' - band diminishes



2. Match the following column and select the correct option.

Column I

A. Floating ribs

B. Acromion

C. Scapula

D. Glenoid cavity

Column II

P. Located between second and seventh ribs

Q. Head of the humerus

R. Clavicle

S. Do not connect with the sternum

(1) A-P, B-R, C-Q, D-S

(2) A-R, B-Q, C-S, D-P

(3) A-S, B-R, C-P, D-Q

(4) A-Q, B-S, C-P, D-R



3. Match the columns I and II.

Column I

A. 'F' actins

B. Tropomyosin

C. Troponin

(1) A-R, B-P, C-Q

(2) A-Q, B-R, C-P

(3) A-P, B-R, C-Q

(4) A-Q, B-P, C-R

Column II

P. Run close to the 'F' actins throughout its length

Q. Masks the active binding sites

R. 'G' actins



4. Pelvic girdle consist of

- (1) Ilium and clavicle
- (2) Two coxal bones
- (3) Two acromion bones
- (4) Acetabulum and glenoid cavity



5. Acetabulum is present in ____ joint

- (1) Hip
- (2) Knee
- (3) Elbow
- (4) Shoulder



6. Read the following statements (I-V)

I. The skull is composed of two sets of bones cranial and facial that total to 22 bones

II. Cranial bones are 14 in number

III. Facial region is made up of 8 bones

IV. A single U-shaped bone called hyoid is present at the base of the buccal cavity and it is also included in the skull.

V. Each middle ear contains three tiny bones malleus, incus and stapes, collectively called ear ossicles

How many of the above statements are incorrect for skeletal system of human being

(1) Four

(2) Three

(3) One

(4) Two



7. Statement A : Muscle fibre is a syncytium.

Statement B : The sarcoplasmic reticulum is a store house of calcium.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



8. Fibrous joints are found in

- (1) Sutures
- (2) Pubic symphysis
- (3) Between ribs and sternum
- (4) All of the above



9. Match the following and mark the correct option

Column I

A Fast muscle fibre

B Slow muscle fibre

C Actin filament

D Sarcomere

Column II

P Myoglobin

Q lactic acid

R Contractile unit

S I-band

(1) A-P, B-Q, C-S, D-R

(2) A-Q, B-P, C-R, D-S

(3) A-Q, B-P, C-S, D-R

(4) A-R, B-Q, C-S, D-P



10. Coccygeal bone is formed by the fusion of bones in man

- (1) Twelve vertebrae
- (2) Four vertebrae
- (3) Five vertebrae
- (4) Six vertebrae



11. Select the correct answer

I. The non coordinated movements of cilia in the trachea help us in removing dust particles and some of the foreign substances inhaled alongwith the atmospheric air.

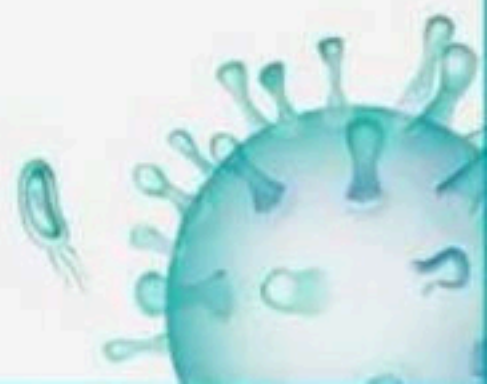
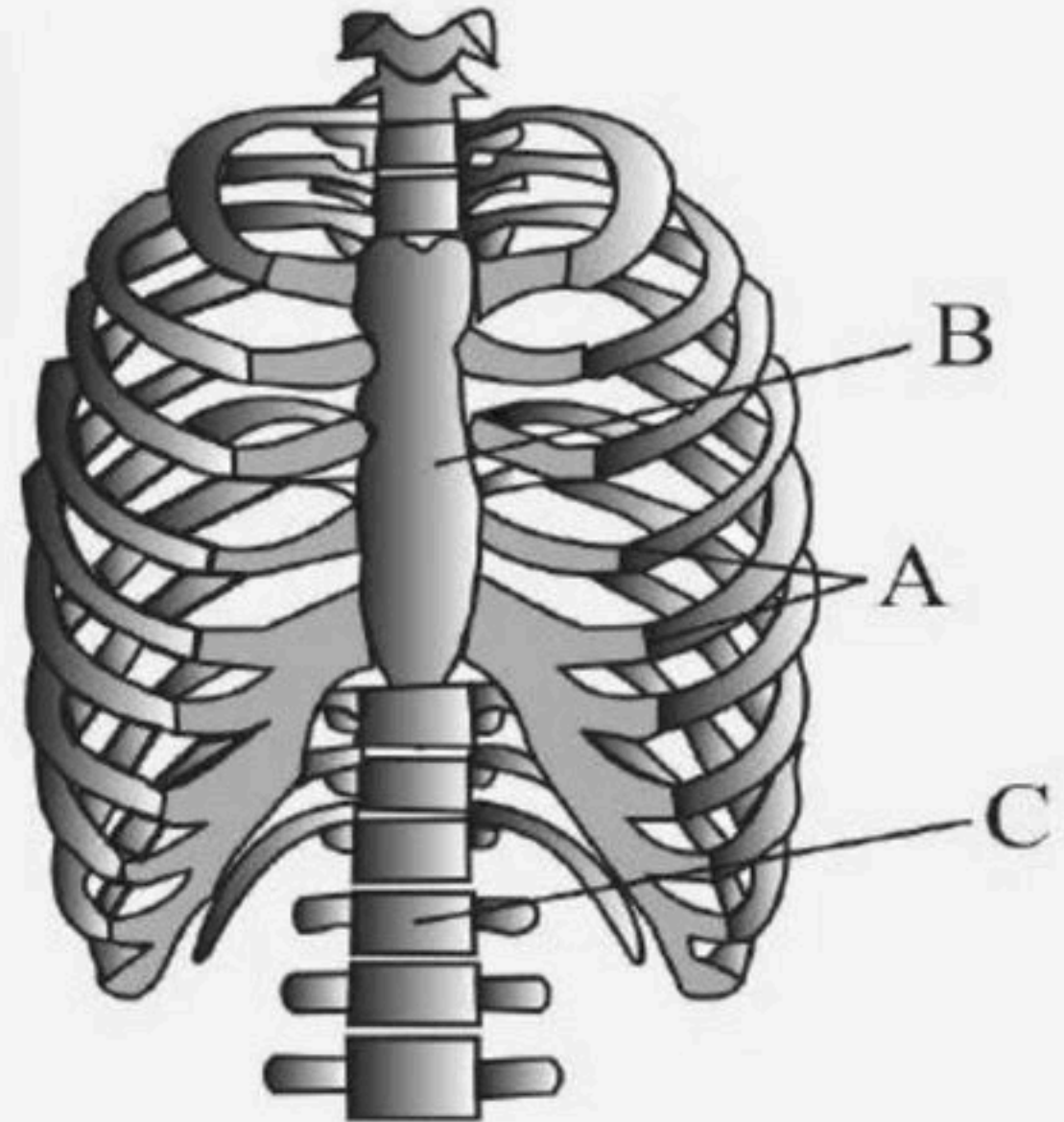
II. Passage of ova through the female reproductive tract is facilitated by the ciliary movement.

- (1) Only I is correct
- (2) Only II is correct
- (3) Both are correct
- (4) None are correct



12. Identify the labeled parts and from the given figure of rib cage.

- | | | |
|--------------------------|----------------------|----------------------|
| (1) A – Ribs | B – Sternum | C – Vertebral column |
| (2) A – Ribs | B – Vertebral column | C – Sternum |
| (3) A – Sternum | B – Ribs | C – Vertebral column |
| (4) A – Vertebral column | B – Ribs | C – Sternum |



13. Calcium ions in the muscle fibre is present in

- (1) Smooth endoplasmic reticulum
- (2) Golgi complex
- (3) Sarcoplasmic reticulum
- (4) Nucleus



14. Find the odd one out

- (1) Arthritis
- (2) Gout
- (3) Tetany
- (4) Osteoarthritis



15. Which of the following muscles are primarily involved in locomotory actions and changes of body postures?

- (1) Skeletal muscles
- (2) Visceral muscles
- (3) Cardiac muscles
- (4) Both (1) and (2)



16. Identify the part that contains

(i) Both actin & myosin

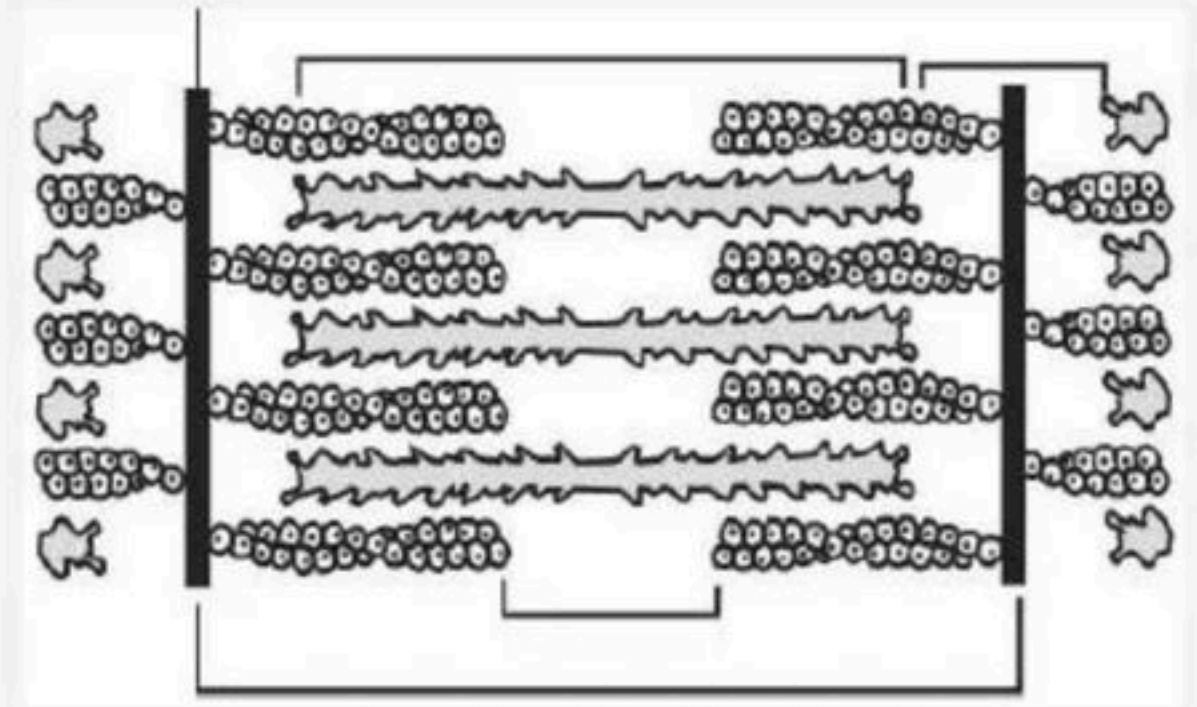
(ii) Area not overlapped by thin filament during rest

(1) (i) A-band (ii) M-line

(2) (i) I-band (ii) H-zone

(3) (i) A-band (ii) M-zone

(4) (i) A-band (ii) H-zone



17. Human appendicular skeleton is made of ____no. of bones

- (1) 80
- (2) 126
- (3) 22
- (4) 29



18. Ribs are attached to

- (1) Scapula
- (2) Sternum
- (3) Clavicle
- (4) Ilium



19. Neural canal is

- (1) The solid portion of vertebrae through which the spinal cord passes
- (2) The hollow portion of vertebrae through which the spinal cord passes
- (3) Both (1) and (2)
- (4) None of these



20. The clavicle or collar bone articulates with

- (1) Cervical vertebrae
- (2) Glenoidal cavity
- (3) Acromion process
- (4) Coxal bone



21. Statement A : Muscle contraction is initiated by signals sent by CNS via a sensory neuron.

Statement B : Repeated activation of the muscles can lead to lactic acid accumulation in them.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



22. The U-shaped bone at the base of the buccal cavity is

- (1) Hyoid
- (2) Ethmoid
- (3) Sphenoid
- (4) Pterygoid



23. Each organised skeletal muscle in our body is made of a number of

- (1) Muscle bundles
- (2) Fascicles
- (3) Both 1 and 2
- (4) Sarcomere



24. Mark the incorrect match

- (1) Saddle joint - Between carpal and metacarpal of thumb
- (2) Ball and socket joint - Between humerus and pectoral girdle
- (3) Cartilagenous joint - Between carpals
- (4) Gliding joint - Between carpals



25. Functional unit of striated muscle is

- (1) I Band
- (2) line
- (3) Myofilament
- (4) Sarcomere



26. Choose the incorrect pair.

- (1) Globular head of meromyosin - Active ATPase enzyme
- (2) Thin elastic membrane holding -M-line thick filaments in A-band
- (3) Dark bands - Anisotropic band
- (4) Each actin (thin) filament is made of two 'F' (filamentous) actins



27. Which of the following is an example of appendicular skeleton?

- (1) Bones of skull
- (2) Bones of vertebral column
- (3) Ribs
- (4) Pectoral and pelvic girdle



28. The fascicles are held together by connective tissue called

- (1) Intercalated disc
- (2) Myofibril
- (3) Fascia
- (4) All of these



29. Calcium is important in skeletal muscle contraction because it is needed for

- (1) Unmasking of actin-binding sites on myosin
- (2) Hydrolysis of ATP by myosin head
- (3) Unmasking of myosin-binding sites on actin
- (4) Polymerization of globular actins into filamentous actin



30. Match the columns I and II.

Column I		Column II	
A	Myasthenia gravis	P	Rapid spasms
B	Tetany	Q	Accumulation of uric acid crystals
C	Gout	R	Auto immune disorder

(1) A-R, B-P, C-Q

(2) A-Q, B-R, C-P

(3) A-P, B-R, C-Q

(4) A-Q, B-P, C-R



31. Match the columns I and II.

Column I		Column II	
A	Bone	P	Chondroitin salts
B	Cartilage	Q	Aerobic muscles
C	Red fibre	R	Calcium salts
D	White fibres	S	Anarobic muscles

(1) A-R, B-P, C-Q, D-S

(2) A-S, B-R, C-P, D-Q

(3) A-P, B-R, C-Q, D-S

(4) A-Q, B-P, C-R, D-S



32. Match the columns I and II.

Column I		Column II	
A	Sarcolemma	P	Alternate dark and light bands
B	Sarcoplasmic reticulum	Q	Plasma membrane
C	Myofibrils	R	Store house of calcium

(1) A-R, B-P, C-Q

(2) A-Q, B-R, C-P

(3) A-P, B-R, C-Q

(4) A-Q, B-P, C-R



33. Match the columns I and II.

Column I		Column II	
A	Ball and socket joint	P	Femur and hip bone
B	Pivot joint	Q	Knee joint
C	Hinge joint	R	Between atlas and axis
D	Saddle joint	S	Between carpal and metacarpal of thumb

(1) A–R, B–P, C–Q, D–S

(2) A–S, B–R, C–P, D–Q

(3) A–P, B–R, C–Q, D–S

(4) A–Q, B–P, C–R, D–S



34. Match the columns I and II.

Column I		Column II	
A	Fatigue	P	Sarcoplasmic reticulum is high
B	Red fibres	Q	Anaerobic breakdown of glycogen
C	White fibres	R	Plenty of mitochondria

(1) A-R, B-P, C-Q

(2) A-Q, B-R, C-P

(3) A-P, B-R, C-Q

(4) A-Q, B-P, C-R



35. Match the columns I and II, and choose the correct combination from the options given

Column I		Column II	
A	Smooth muscle	P	Myoglobin
B	Tropomyosin	Q	Thin filament
C	Red muscle	R	Sutures
D	Skull	S	Involuntary

(1) A-S, B-Q, C-P, D-R

(2) A-S, B-P, C-Q, D-R

(3) A-P, B-Q, C-S, D-R

(4) A-Q, B-R, C-S, D-P



36. Match the columns I and II.

Column I		Column II	
A	Osteoporosis	P	Progressive degeneration of skeletal muscle
B	Arthritis	Q	Decreased levels of estrogen
C	Muscular dystrophy	R	Inflammation of joints

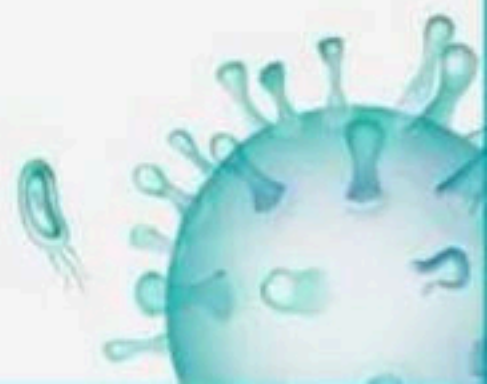
- (1) A – R, B – P, C – Q
- (2) A – Q, B – R, C – P
- (3) A – P, B – R, C – Q
- (4) A – Q, B – P, C – R



37. Match the columns I and II

Column I		Column II	
A	Fibrous joint	P	Between humerus and pectoral girdle
B	Cartilaginous joint	Q	Sutures
C	Synovial joint	R	Between two adjacent vertebrae

- (1) A – R , B – P , C – Q
- (2) A – Q , B – R , C – P
- (3) A – P , B – R , C – Q
- (4) A – Q , B – P , C – R





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ACTION AGENDA

HOME (/HOME) > ACTION AGENDA () > DEVELOPMENT OF TESTS

ACTION AGENDA

➊ Research(/Research)
➋ Internship, Fellowship and Visiting Scholar Programs in Research(/Internship)
➌ Development of Tests and Test Items(/DevelopmentofTests)
➍ Test Administration(/TestAdministration)
➎ Test Markingand Reporting(/TestMarking)
➏ Empowering Boards of School Education to bridge the Gap Between Board Exams and Entrance Examinations (/Bridging)
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DEVELOPMENT OF TESTS AND TEST ITEMS

NTA will develop tests to improve quality in education by administering research based valid, reliable, efficient, transparent, fair and international level assessments. The best subject matter experts, psychometricians and IT delivery and security professionals will ensure that the current gaps in existing assessment systems are properly identified and bridged. Professionals including test specialists, test reviewers, editors, teachers and specialists in the subject or skill being tested — will be involved by NTA in developing "test items." And that is why all questions (or "items") are put through multiple, rigorous reviews and meet the highest standards for quality and fairness in testing.

The tests developed by NTA will have the following special features –

A) TEST ITEMS BEING CREATED BY SUBJECT EXPERTS IN COLLABORATION WITH PSYCHOMETRICIANS

The tests will be jointly developed by Subject matter experts and psychometricians. The data of every question of every test will be statistically analyzed as well as put through psychometric analysis. These aspects will be very critical in terms of ensuring that the tests are well distributed. Attempt will be made to ensure that the present situation of a very low mean and skewed distribution does not exist. Students will also be provided with some simulated mock tests so that they have an idea of how to prepare for the CBT.

B) CREATION OF QUESTION/ITEM BANKS /QUESTION BANKS

Furthermore, as a next step NTA will create a large size properly indexed question bank for every subject. The question papers will be prepared by selecting items randomly from the question banks based on a scientifically designed blueprint and algorithm of the paper. These question banks will be created using the modern technology of Artificial Intelligence and computer software. The main objective of using such an approach will be for generation of a large number of tests in a short period of time so that NTA delivered tests can be held multiple times in a single calendar year without compromising on rigor of test building test items and the sanctity of the test.

Overview of the key steps NTA will take when developing a new test-

Step 1: Determining Test Objectives to match with needs

The commissioning organisations or directives of the government will identify a need to measure certain skills or knowledge. Once a decision is made to develop a test to accommodate this need, test developers will address the following:

- Who will take the test and for what purpose?
- What skills and/or areas of knowledge will comprise the construct to be measured?
- How will test takers be able to use their knowledge?
- What kind of blueprint / table of specifications / algorithm will be needed to measure the intended construct?
- How long should the test be?
- How difficult should the test be?

Step 2: Test Development Committees

The answers for the questions in Step 1 are usually completed with the help of test development committees, which typically will consist of educators, and/or other professionals appointed by NTA with the guidance of the stakeholder organisation. Responsibilities of these test development committees (one for each of the tests conducted by NTA) may include:

- defining test objectives and specifications
- helping ensure test questions are unbiased
- determining test format (e.g., multiple-choice, essay, constructed-response, etc.)
- considering supplemental test materials

Step 3: Item Writing Committees

For each of the subject area of a test, there will be an item writing committee of not less than 30 item writers. They will work on item writing with the help of psychometricians and statisticians. Their responsibilities will include:

- reviewing test questions, or test items, given by NTA staff
- rewriting / moderating / refining (Tried Exercise) test questions

Step 4: Moderating and Writing Questions

Each test question — written by NTA staff or item writing committees — undergoes numerous reviews and revisions to ensure it is as clear as possible, that it has only one correct answer among the options provided on

the test and that it conforms to the style rules used throughout the test. Marking schemes and scoring rubrics for open-ended responses (constructed response item), such as short written answers and essays, go through similar reviews.

Step 5: Detecting and Removing Unfair Questions

To meet the Standards for Quality and Fairness guidelines, trained reviewers will then critically examine and evaluate each individual test question, the test as a whole to ensure that language, symbols, words, phrases and content generally regarded as sexist, racist or otherwise inappropriate or offensive to any subgroup of the test-taking population are eliminated. A rigorous checklist in this regard will be co-developed by NTA. NTA statisticians also will identify questions on which two groups of test takers who have demonstrated similar knowledge or skills perform differently on the test through a process called Differential Item Functioning (DIF) based on data of previous years' administration. If one group performs consistently better than another on a particular question, that question may be deemed biased or unsatisfactory.

Note: If people in different groups actually differ in their average levels of relevant knowledge or skills, a fair test question will reflect those differences.

An insight from every test conducted henceforth by the NTA will be used to improve the functioning of items and tests for further administration

Step 6: Assembling the Test

After the test is assembled, it is reviewed by other specialists, committee members and sometimes other outside experts. Each reviewer answers all questions independently and submits a list of correct answers to the test developers. The lists are compared with the NTA answer key to verify that the intended answer is, indeed, the correct answer. Any discrepancies are resolved before the test is published.

Step 7: Making Sure – Even After the Test is Administered – that the Test Questions are Functioning Properly

Even after the test has been administered, statisticians and test developers will preliminary conduct a key check analysis and review to make sure that test questions are working as intended. Before final marking takes place, each question will undergo preliminary statistical analysis and results will be reviewed question by question. If a problem is detected, such as the identification of a misleading answer to a question, corrective action, such as not marking the question, will be taken before final marking and marks reporting takes place. Tests will also be reviewed for comparability across various test forms. Performance on one version form of the test should reasonably predict performance on any other form/version of the test. If comparability is high, results will be similar no matter which version a test taker completes.

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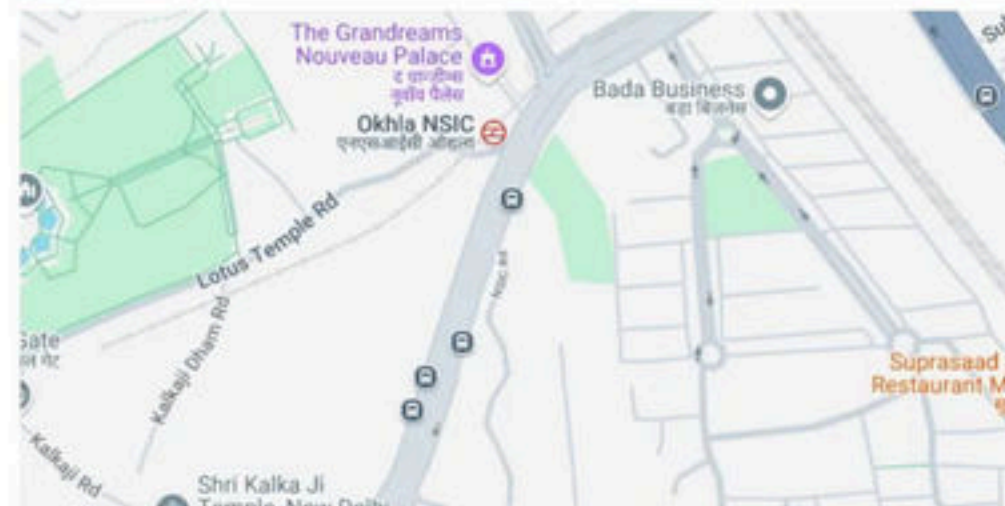
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38. Match the columns I and II.

Column I		Column II	
A	Amoeboid movement	P	Movement of our limbs
B	Ciliary movement	Q	Spermatozoa
C	Muscular movement	R	Trachea
D	Flagellar movement	S	Macrophages and leucocytes

- (1) A–R, B–P, C–Q, D–S
- (2) A–S, B–R, C–P, D–Q
- (3) A–P, B–R, C–Q, D–S
- (4) A–Q, B–P, C–R, D–S



39. Match the columns I and II.

Column I		Column II	
A	Centre of each 'I' band	P	'Z' line
B	Centre of each 'A' band	Q	'H' zone
C	Not overlapped by thin filaments	R	'M' line

- (1) A – R, B – P, C – Q
- (2) A – Q, B – R, C – P
- (3) A – P, B – R, C – Q
- (4) A – Q, B – P, C – R



40. Match Column I with Column II :

Choose the correct answer from the options given below

Column I		Column II	
A	Fibrous joints	P	Adjacent vertebrae, limited movement
B	Cartilaginous joints	Q	Humerus and Pectoral girdle, rotational movement
C	Hinge joints	R	Skull, don't allow any movement
D	Ball and socket joints	S	Knee, help in locomotion

(1) A – S, B – Q, C – R, D – P

(2) A – P, B – R, C – Q, D – S

(3) A – Q, B – R, C – P, D – S

(4) A – R, B – P, C – S, D – Q



41. Match the columns I and II.

Column I		Column II	
A	Parietal bone	P	Upper jaw
B	Maxilla	Q	Cranium
C	Mandible	R	Lower jaw
D	Hyoid	S	U-shaped

(1) A–R, B–P, C–Q, D–S

(2) A–S, B–R, C–P, D–Q

(3) A–P, B–R, C–Q, D–S

(4) A–Q, B–P, C–R, D–S



42. Match the columns I and II.

Column I		Column II	
A	True ribs	P	8th, 9th, 10th rib pair
B	False ribs	Q	First seven pairs
C	Floating ribs	R	11th & 12th rib pair

(1) A–R, B–P, C–Q

(2) A–Q, B–R, C–P

(3) A–P, B–R, C–Q

(4) A–Q, B–P, C–R



43. Match the followings and mark the correct option

Column I		Column II	
A	Sternum	P	Synovial fluid
B	Glenoid cavity	Q	Vertebrae
C	Freely movable joint	R	Pectoral girdle
D	Cartilaginous joint	S	Flat bones

(1) A-Q, B-P, C-R, D-S

(2) A-S, B-R, C-P, D-Q

(3) A-Q, B-P, C-S, D-R

(4) A-S, B-P, C-Q, D-S



44. Match the columns I and II

Column I		Column II	
A	Cervical	P	12
B	Thoracic	Q	5
C	Lumbar	R	7
D	Sacral	S	1-fused/4
E	Coccygeal	T	1-fused/5

(1) A-R , B-P , C-Q, D-T, E-S

(2) A-S , B-R , C-P, D-Q , E-T

(3) A-P, B-R, C-Q, D-S, E-T

(4) A-Q, B-P, C-R, D-S, E-T



45. Match Column - I with Column - II.

Choose the correct answer from the options given below.

Column I		Column II	
A	Scapula	P	Cartilaginous joints
B	Cranium	Q	Flat bone
C	Sternum	R	Fibrous joints
D	Vertebral column	S	Triangular flat bone

(1) A-P, B-R, C-Q, D-S

(2) A-Q, B-R, C-S, D-P

(3) A-S, B-Q, C-R, D-P

(4) A-S, B-R, C-Q, D-P



46. Match the columns I and II.

Column I

Column II

A. Carpals

P. 14

B. Metacarpals

Q. 8

C. Phalanges

R. 5

(1) A – R , B – P , C – Q

(2) A – Q , B – R , C – P

(3) A – P , B – R , C – Q

(4) A – Q , B – P , C – R



47. Match Column I with Column II

Column I		Column II	
A	Cartilaginous Joint	P	Between flat skull bones
B	Ball	Q	Between adjacent vertebrae in vertebral column
C	Fibrous Joint	R	Between carpal and Socket Joint bones
D	Saddle Joint	S	Between Humerus and Pectoral girdle

(1) A-R, B-P, C-Q, D-S

(2) A-Q, B-S, C-P, D-R

(3) A-P, B-S, C-R, D-Q

(4) A-Q, B-S, C-R, D-P



48. Match the columns I and II?

Column I		Column II	
A	Skeletal muscles	P	Nonstriated muscle
B	Smooth muscles	Q	Voluntary muscle
C	Cardiac muscles	R	Striated but involuntary

(1) A-R, B-P, C-Q

(2) A-Q, B-R, C-P

(3) A-P, B-R, C-Q

(4) A-Q, B-P, C-R



49. Match the columns I and II.

Column I		Column II	
A	Cranial bones	P	8
B	Facial bones	Q	6
C	Ear ossicles	R	14
D	Hyoid	S	1

- (1) A – R, B – P, C – Q, D – S
- (2) A – S, B – R, C – P, D – Q
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – P, C – R, D – S



50. Statement A : Cardiac muscles are the muscles present in the myocardium.

Statement B : Cardiac muscles are controlled by nervous system directly.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



51. Which of the following statements about the molecular arrangement of actin and myosin in myofibrils is false?

I. Each actin (thin filament) is made of F (filamentous) actins

II. F-actin is the polymer of (globular) actin.

III. 2F-actins are twisted into a helix.

IV. Two strands of tropomyosin (protein) lie in the grooves of F-actin.

V. Troponin molecules (complex proteins) are distributed at regular intervals on the tropomyosin

VI. Troponin forms the head of the myosin molecule

VII. The myosin is a polymerised protein

(1) I, III, VII

(2) Only VII

(3) Only III

(4) VI only



52. Statement A : About of the body weight is contributed by muscles

Statement B : Muscles have special properties like excitability, connectivity and extensibility

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



53. Read the following I to IV statements and select the one option that contains both correct statements

I. Z-line is present in the centre of the light band.

II. Thin filaments are firmly attached to the M-line.

III. The central part of thick filaments, not overlapped by thin filaments is called Z-band

IV. Light band contains only thin filaments

(1) II and III

(2) I and IV

(3) I and III

(4) II and IV



54. Read the following statements and find out the incorrect statement

- I. Hyoid is present at the roof of the buccal cavity and include in the skull.**
- II. The skull region articulates with the superior region of the vertebral column with the help of two occipital condyles**
- III. First vertebra is the axis which articulates with the occipital condyles.**
- IV. Sternum is a flat bone in cranium, face, hyoid and middle ear are respectively**

(1) II, III and V

(2) III, IV and V

(3) I, III and IV

(4) I, II and IV



55. Statement A : Voluntary movements in organism help in locomotion

Statement B : Dust particles are removed from trachea due to the presence of villi on the epithelium on inner side

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



56. Read the statements regarding muscle protein

I. Actin is a thin filament and is made up of two Factins.

II. Complex protein, tropomyosin is distributed at regular intervals on the troponin.

III. Myosin is a thick filament which is also a polymerized protein.

IV. Globular head of meromyosin consists of light meromysin (LMM).

Of the above statements:

(1) I, III and IV are correct (2) I, II and III are correct

(3) I and III are correct (4) I, II and IV are correct



57. Statement A : The tentacles of Hydra are locomotory in function

Statement B : In Paramecium, flagella help in the movement of food through cytopharynx

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



58. Pick the true statements with reference to human beings

I. Neural canal is a cavity in which spinal cord runs.

II. Scapula has hook like Coronoid process.

III. Imperfect joints are present in sternum- ribs-joints.

IV. Number of vertebrae in adult is 26 .

(1) II, III and IV

(2) I, III and IV

(3) I, II and III

(4) II and IV



59. Statement A : The most abundant joints in human body are synovial joints

Statement B : Muscles are specialised tissues of mesodermal origin

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



60. Statement A : The myosin rich zone called A-band maintains its length during sarcomere contraction.

Statement B : The length of I-band remains the same during muscle contraction

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



61. Statement A : Each half of pelvic girdle consists of a clavicle and scapula

Statement B : Clavicle is a long slender bone with three curvatures

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



62. Consider the following four statements and select the correct options

I. Actin is present in thin filament

II. Sternum is present on ventral side of the body

III. There are 11 pairs of ribs in man

IV. H-zone of striated muscle fibre represents both thick and thin filaments

	I	II	III	IV
(1)	T	F	F	T
(2)	T	T	F	F
(3)	F	T	F	T
(4)	F	F	T	T



63. Statement A : Floating ribs are not connected to sternum ventrally

Statement B : Human ribs are bicephalic as they possess two articulation surfaces.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



64. Statement A : Each myofibril consists of alternate white and dark bands.

Statement B : In muscle fibre, F-actin and tropomyosin are helically wound filaments

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



65. Statement A : Knee joint is the hinge type of joint.

Statement B : Knee joint is the gliding type of joint.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



66. Statement A : Skeletal muscles activities are under the voluntary control of the nervous system

Statement B : Skeletal muscles are primarily involved in locomotory actions and changes of body postures

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



67. Statement A : Muscle contraction does not need ATP.

Statement B : Muscle fatigue occurs due to pyruvic acid formation.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



68. Statement A : Cilia are seen in Paramecium and other ciliates

Statement B : Cilia are present in trachea, duodenum, vasa efferentia and oviducts

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



69. Statement A : Axial skeleton comprises 80 bones

Statement B : Skull has total 22 bones in cranial and facial region including hyoid

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



70. Study the following statements and select the correct option

I. The accumulation of lactic acid in the muscle causes fatigue

II. The phosphogen in the vertebrate muscle is arginine phosphate

III. Cori's cycle occurs in the muscle

IV. ATP is resynthesised in the muscle by the phosphorylation of ADP by a phosphogen

(1) I and II

(2) II and III

(3) III and IV

(4) I and IV



71. Statement A : Acetylcholine is released when the neural signal reaches to the motor end-plate.

Statement B : During muscle contraction, isotropic bands get elongated.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



72. Statement A : Amount of myoglobin is high in red muscle fibres

Statement B : Pseudopodial movements as in Amoeba are shown by cytoskeletal elements like microfilaments

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



73. Statement A : The cranium and tarsals are parts of axial skeleton

Statement B : Both tarsals and cranium have 8 bones each

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



74. Statement A : Human skull is dicondylic.

Statement B : It is attached to vertebral column through two occipital condyles

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



75. Statement A : Ciliary movements are observed in internal tubular organs.

Statement B : All muscle bundles contain a covering called fascia

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



76. Which of the following statements are incorrect regarding a normal human?

I. Scapula is a large triangular flat bone, situated on the ventral side of the thorax.

II. The pelvic girdle has two coxal bones.

III. Patella is a cup shaped bone covering the elbow joint.

IV. Metacarpals are five in number

(1) I and III

(2) II and IV

(3) I and II

(4) II and III



77. Which of the following statements are correct regarding skeletal muscle?

I. Muscle bundles are held together by collagenous connective tissue layer called fascicle.

II. Sarcoplasmic reticulum of muscle fibre is a store house of calcium ions.

III. Striated appearance of skeletal muscle fibre is due to distribution pattern of actin and myosin proteins.

IV. M line is considered as functional unit of contraction called sarcomere.

Choose the most appropriate answer from the options given below:

(1) I, II and III only

(2) II and III only

(3) I, III and IV only

(4) III and IV only



78. Statement A : Each myosin is a polymerised protein, many meromyosins constitute one thin filament (myosin).

Statement B : The globular head of actin is an active sites for myosin .

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



79. Statement A : All movements are locomotions, but all locomotions are not movements.

Statement B : Streaming of protoplasm in Amoeba is a type of amoeboid movement

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



80. Statement A : Smooth muscles are known as unstriated muscles

Statement B : They are found close to skeletal components of body like bones

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



81. Statement A : On stimulation, a muscle cell releases calcium ions (Ca^{2+}) from sarcoplasmic reticulum

Statement B : By reacting with a protein complex, Ca^{2+} uncover active sites on the myosin filaments.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



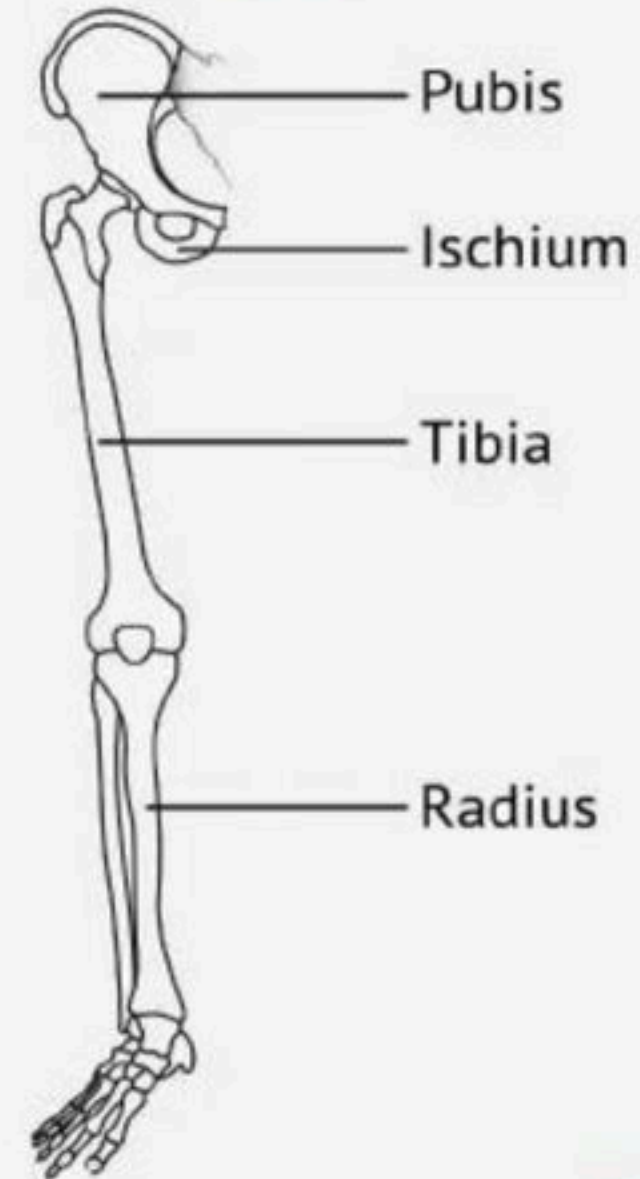
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graph TD; A[Skeletal system] --> B[Axial skeleton]; A --> C[A]; B --> D[Skull]; B --> E[B]; B --> F[Sternum]; B --> G[C]; C --> H[Limbs]; C --> I[D]
```

The flowchart classifies the Skeletal system into two main branches: Axial skeleton and A. The Axial skeleton further branches into Skull, B, Sternum, and C. Branch A branches into Limbs and D.

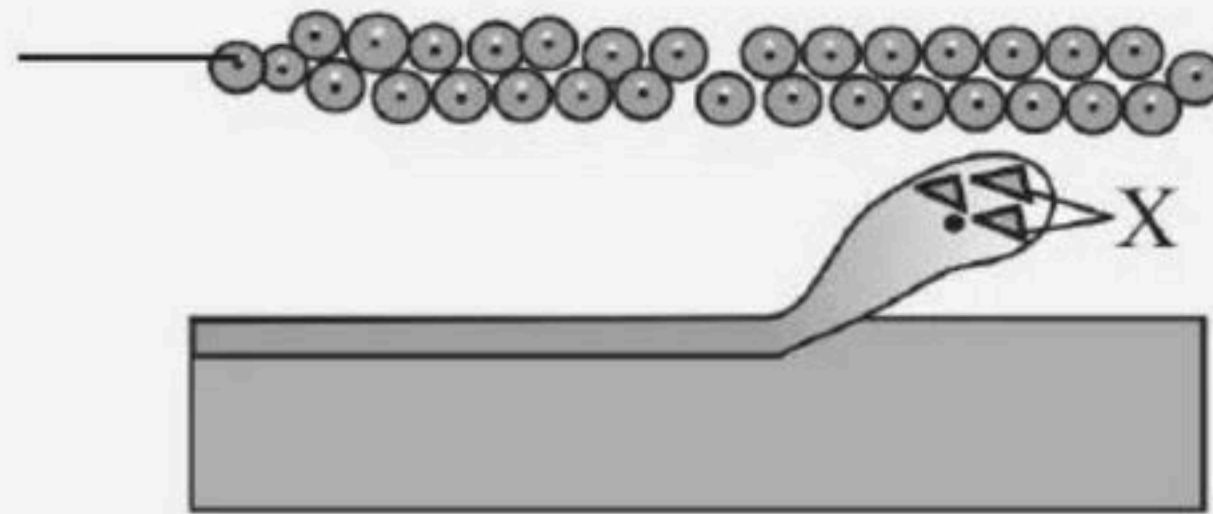
- (1) A appendicular skeleton B cranium, C facial bones, D pelvis
- (2) A girdles, B Vertebral column, C cranium, D coxal bone
- (3) A appendicular skeleton, B cranium bones, C facial bones, D girdles
- (4) A appendicular skeleton, B vertebral column, C ribs, D girdles

83. Identify the correctly labelled bone in the following diagram.

- (1) Tibia
- (2) Radius
- (3) Pubis
- (4) Ischium



84. Identify 'X' in the following

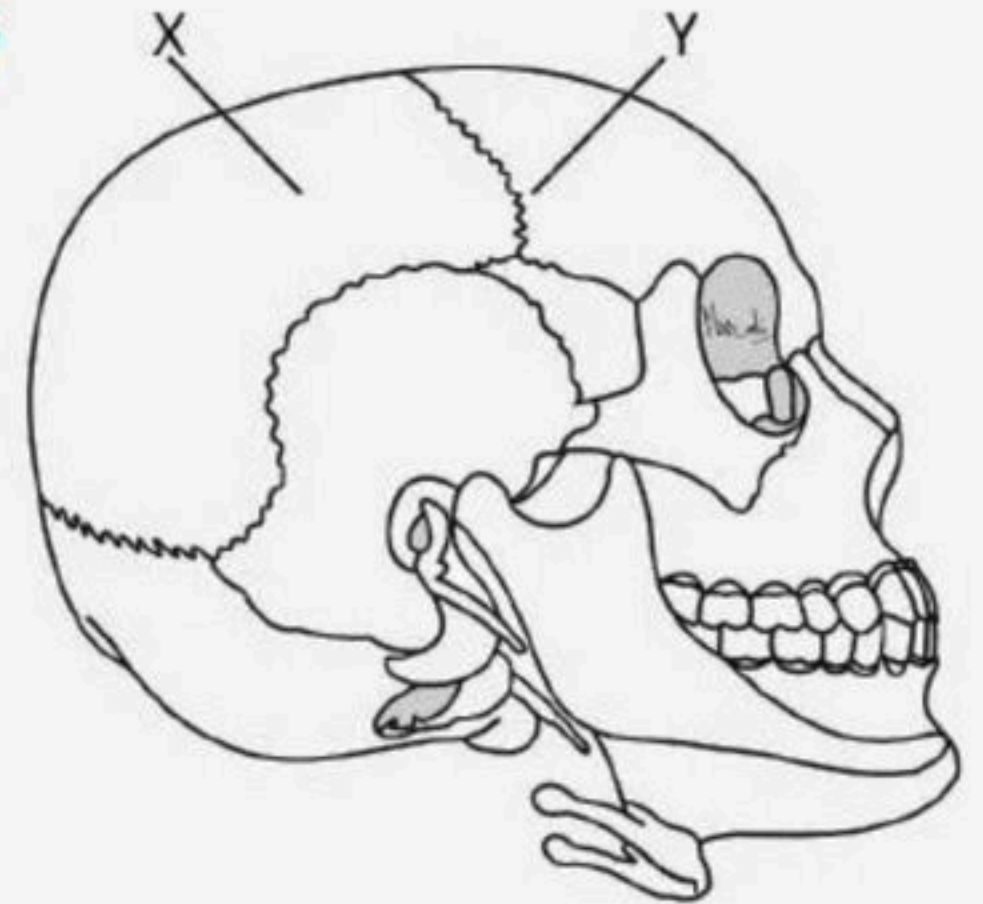


- (1) ATP
- (2) ADP + P
- (3) ATP + P
- (4) APP + D



85. Select the option that correctly identifies the bones labelled X and Y in the following diagram.

X	Y
(1) Frontal	Parietal
(2) Occipital	Frontal
(3) Parietal	Frontal
(4) Parietal	Temporal



86. Three types of muscles are given as and Identify the correct matching pair along with their location in human body:

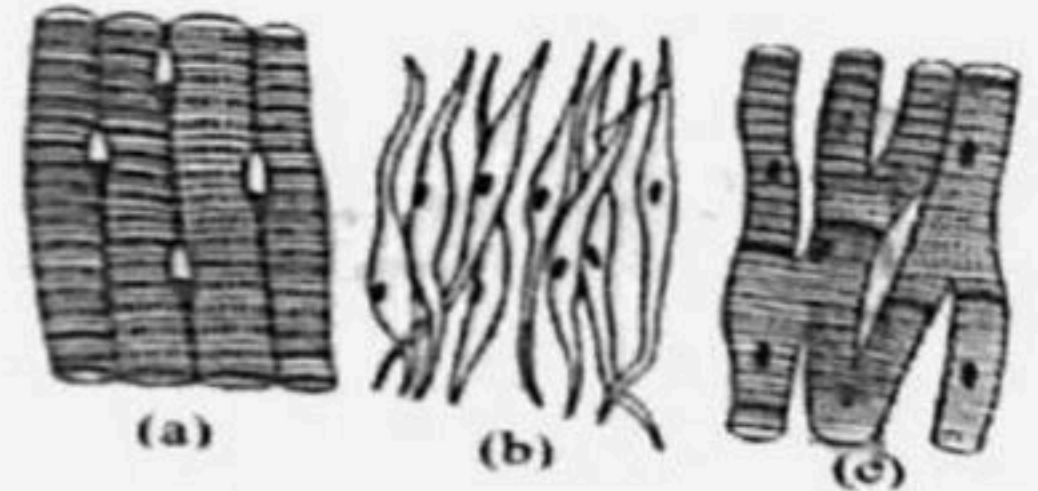
Name of Muscle/location

(1) (a) Smooth - Toes (b) Skeletal - Legs (c) Cardiac - Heart

(2) (a) Skeletal - Triceps (b) Smooth - Stomach (c) Cardiac - Heart

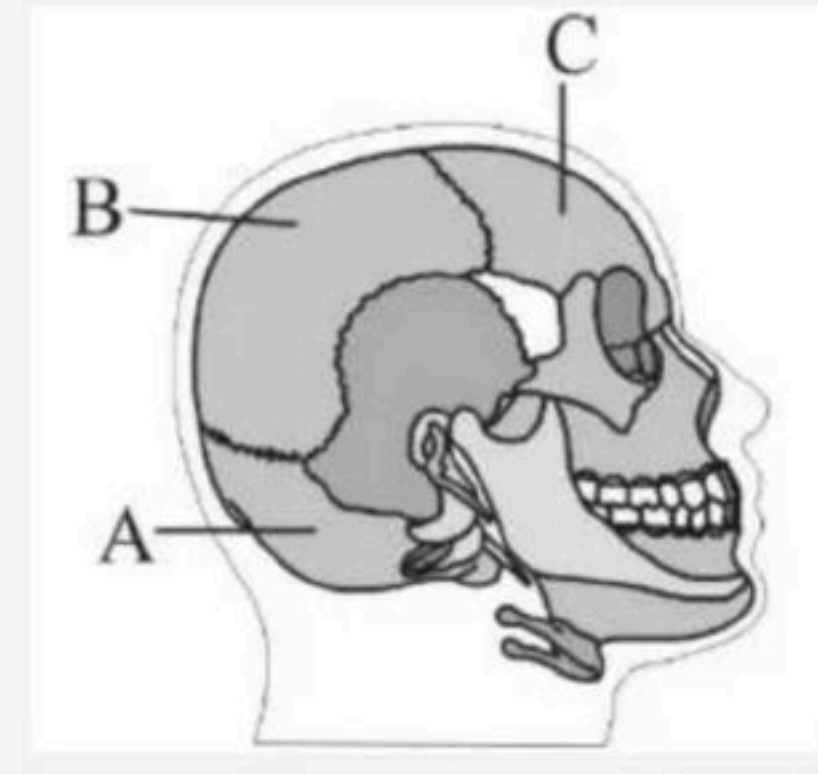
(3) (a) Skeletal - Biceps (b) Involuntary - Intestine (c) Smooth - Heart

(4) (a) Involuntary - Nose tip (b) Skeletal - Bone (c) Cardiac - Heart



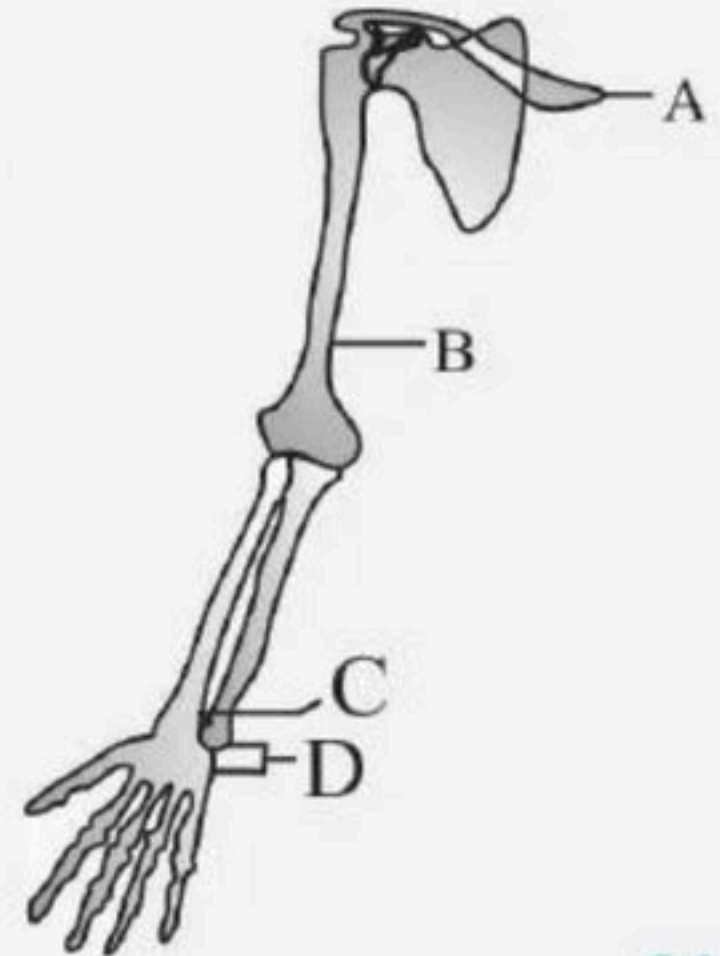
87. Identify A, B and C in the diagram

- (1) A-Occipital bone, B-Frontal bone, C-Parietal bone
- (2) A-Occipital bone, B-Parietal bone, C-Frontal bone
- (3) A-Temporal bone, B-Parietal bone, C-Frontal bone
- (4) A-Temporal bone, B-Frontal bone, C-Parietal bone



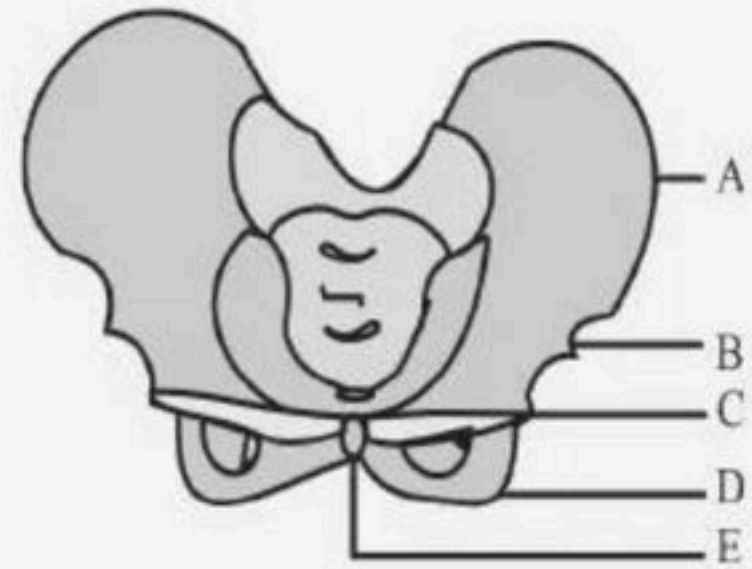
88. Examine the figure of pectoral girdle and forelimb. Identify the parts A, B, C and D.

- | | | | | |
|-----|--------------|-------------|------------|---------------|
| (1) | A – Clavicle | B – Femur | C – Radius | D – Carpals |
| (2) | A – Scapula | B – Humerus | C – Ulna | D – Carpals |
| (3) | A – Clavicle | B – Humerus | C – Radius | D – Carpals |
| (4) | A – Scapula | B – Humerus | C – Ulna | D – Phalanges |

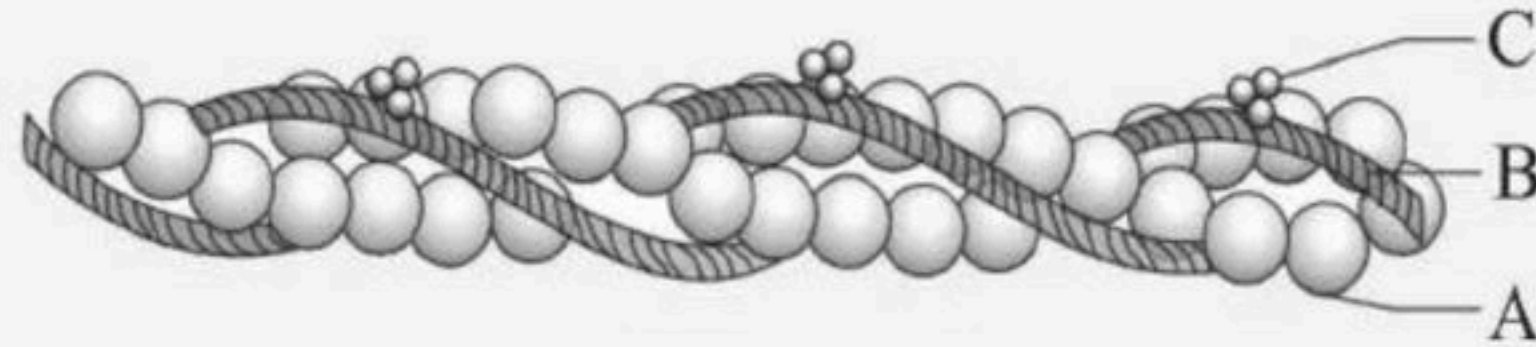


89. In the pelvic girdle of man A, B, C, D and E respectively represents?

- (1) A-pubis, B-acetabulum, C-ilium, D-ischium, E-pubic symphysis
- (2) A-ilium, B-acetabulum, C-pubis, D-ischium, E-pubic symphysis
- (3) A-ischium, B-acetabulum, C-pubis, D-ilium, E-pubic symphysis
- (4) A-ilium, B-pubis, C-acetabulum, D-pubic symphysis, E-ischium



90. Identify the structure that masks the active binding site for myosin

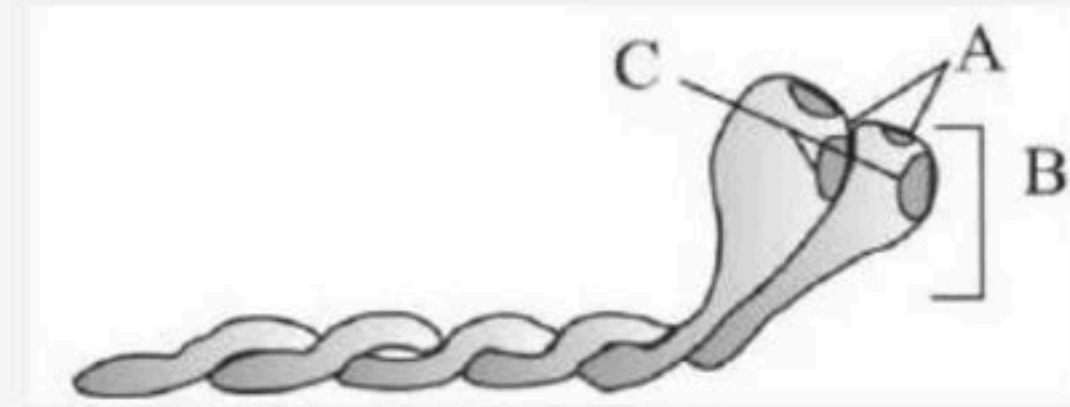


- (1) A
- (2) B
- (3) C
- (4) Myosin itself



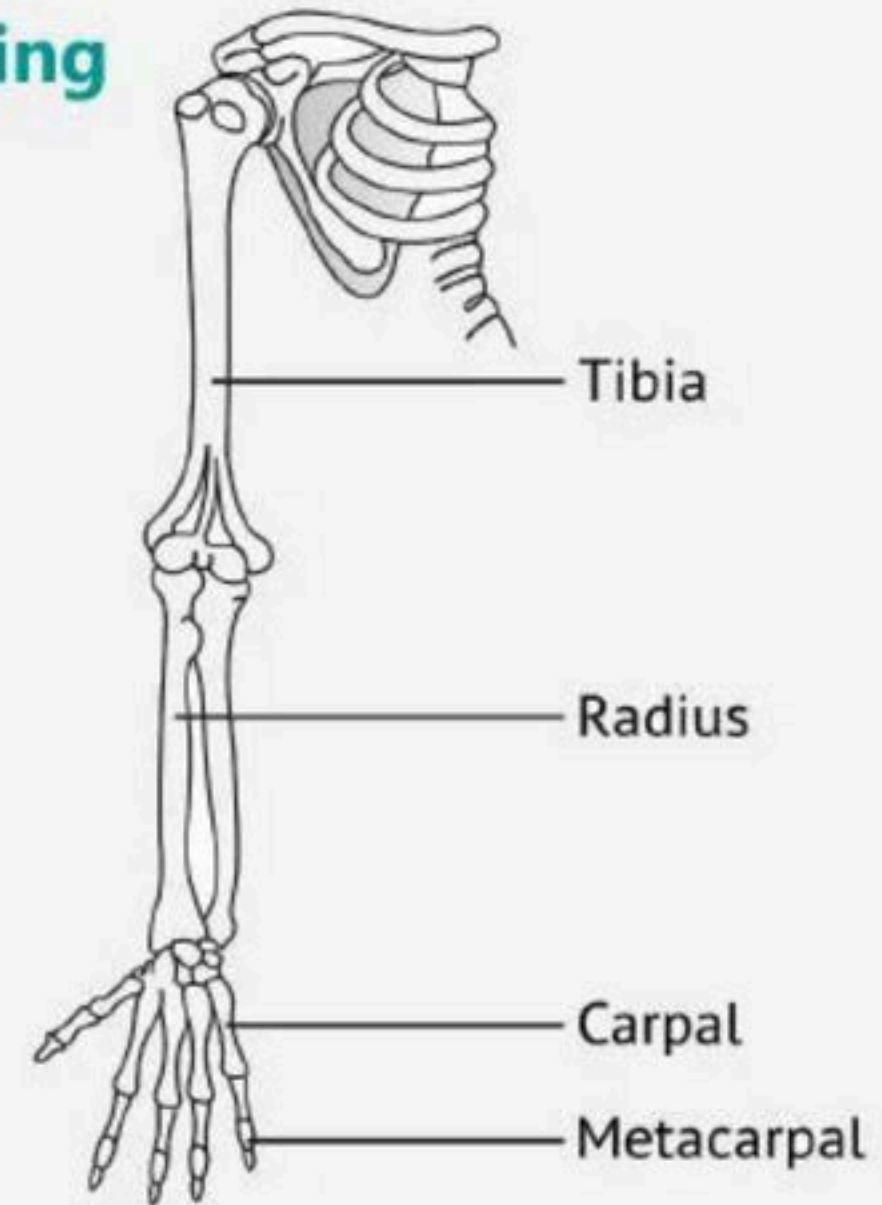
91. Identify the labeled parts from A, B, C from the diagram of myosin monomer

- (1) A-Head B-Cross arm C-ATP binding site
- (2) A-Head B-Cross arm C-GTP binding site
- (3) A-Cross B-Head arm C-ATP binding site
- (4) A-Head B-Cross arm C- binding site



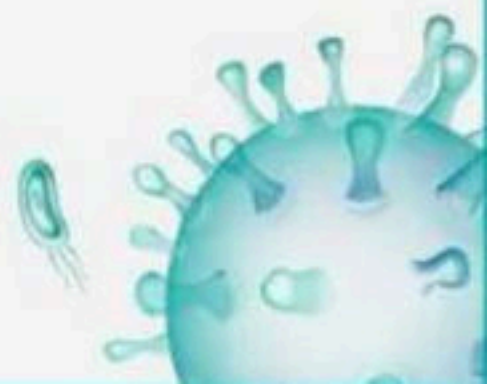
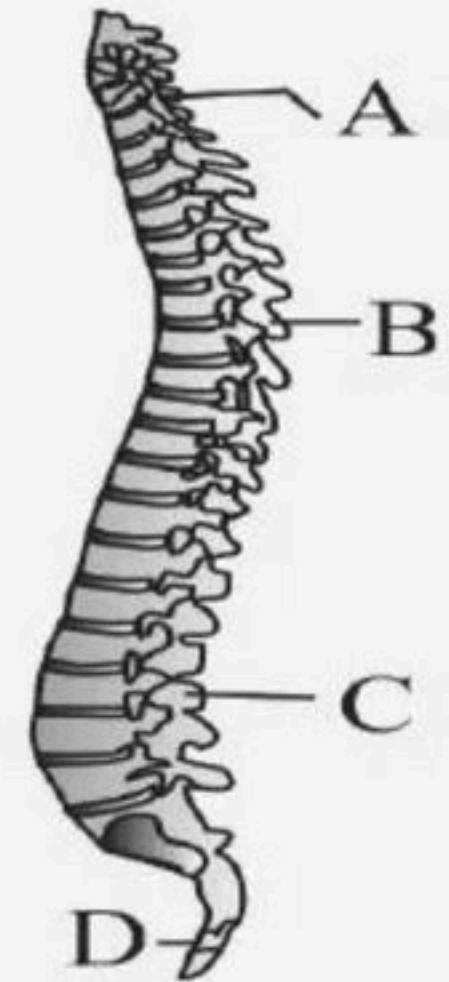
92. Identify the correctly labelled bone in the following diagram.

- (1) Tibia
- (2) Radius
- (3) Carpal
- (4) Metacarpal

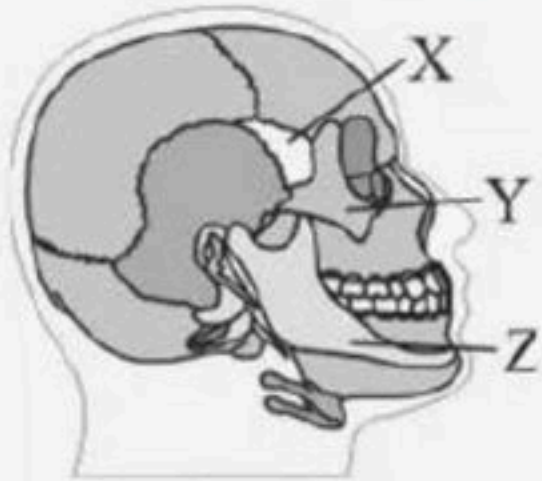


93. Given below is the figure of vertebral column. Identify its parts A, B, C and D.

- | | | | |
|-----|-----------------------|-----------------------|-----------------------|
| (1) | A – Cervical vertebra | B – Thoracic vertebra | C – Lumbar vertebra |
| | D – Coccyx | | |
| (2) | A – Lumbar vertebra | B – Thoracic vertebra | C – Cervical vertebra |
| | D – Coccyx | | |
| (3) | A – Cervical vertebra | B – Thoracic vertebra | C – Lumbar vertebra |
| | D – Sacrum | | |
| (4) | A – Lumbar vertebra | B – Cervical vertebra | C – Thoracic vertebra |
| | D – Sacrum | | |



94. Identify X, Y and Z.

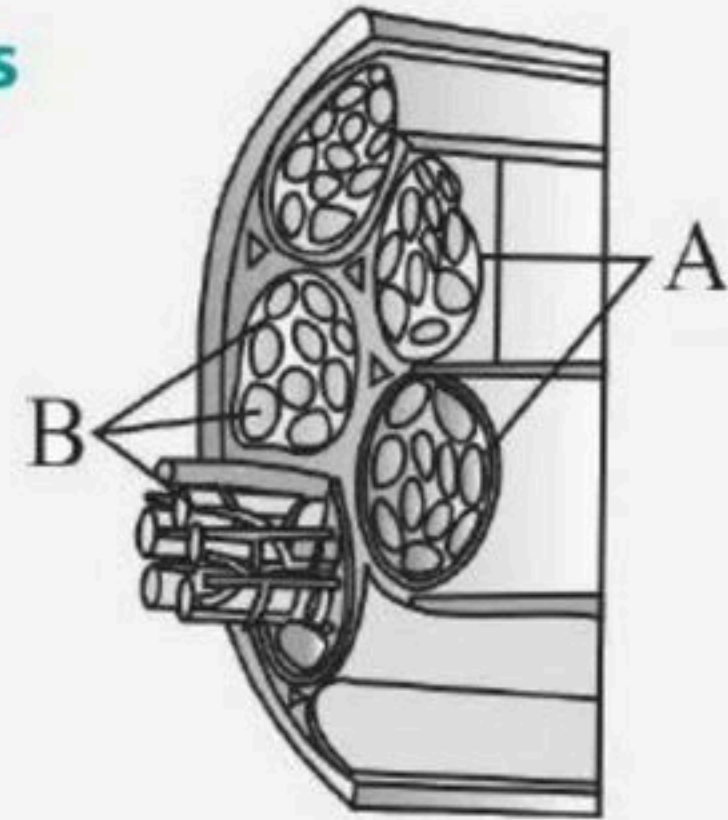


- (1) X-Ethmoid, Y-Nasal, Z-Maxilla
- (2) X-Ethmoid, Y-Zygomatic, Z-Mandible
- (3) X-Sphenoid, Y-Zygomatic, Z-Mandible
- (4) X-Sphenoid, Y-Zygomatic, Z-Maxilla



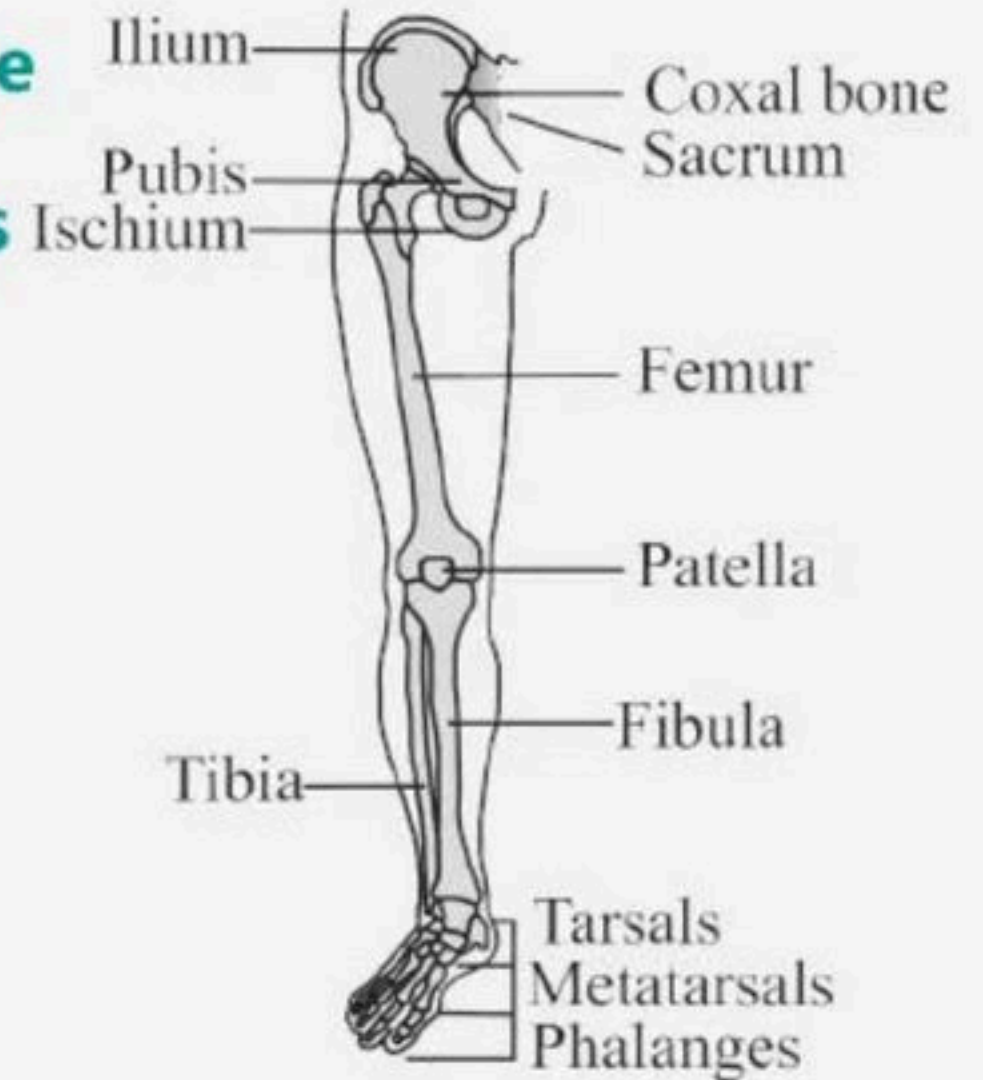
95. Which out of the following represent collagenous

- (1) A - Sarcolemma
- (2) B - Fascia
- (3) A - Fascia
- (4) B - Sarcolemma



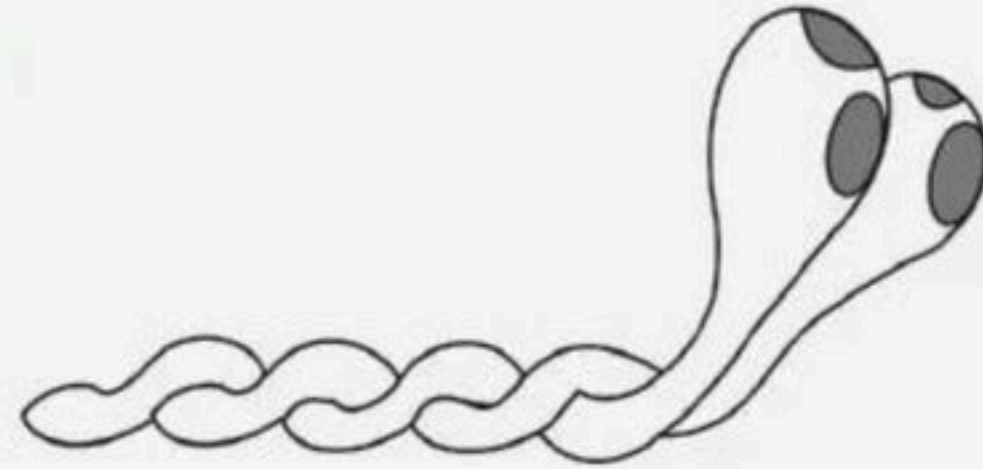
96. Given diagram is a frontal view of the bones of the right pelvic girdle and lower limb. It has two mistakes in labeling. The two incorrectly labeled bones are:

- (1) Ischium and pubis
- (2) Fibula and tibia
- (3) Tarsals and metatarsal
- (4) Sacrum and patella



97. Select the option that correctly identifies the following contractile protein along with its location.

- (1) Meromyosin - Thin filament
- (2) Tropomyosin - Thick filament
- (3) Meromyosin - Thick filament
- (4) Actin - Thin filament



98. Select the option which identifies any one part (A-C) of the following diagram and gives its correct description.

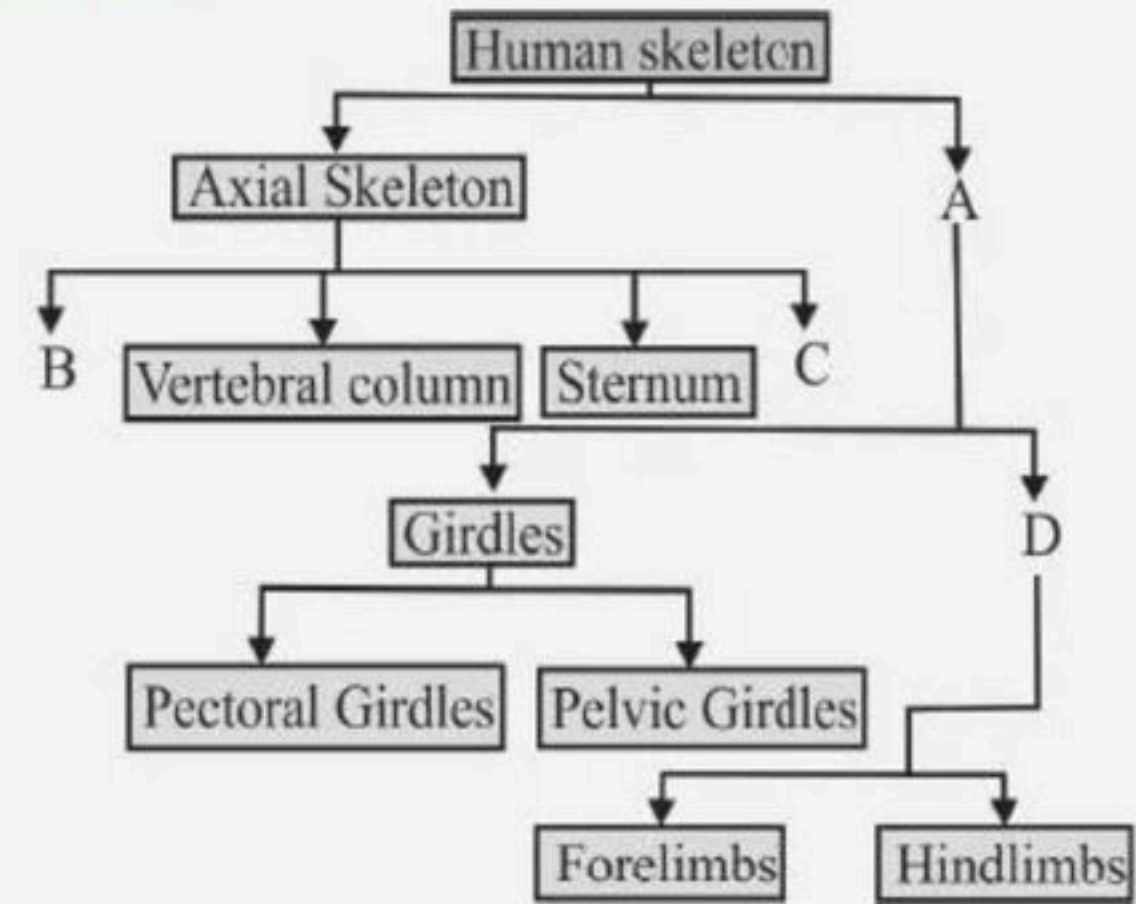


- (1) A: Troponin - A complex protein distributed at regular intervals on the actin
- (2) B: Troponin - Two filaments run close to the F-actins
- (3) C: G-actin - A polymer of monomeric F-actins
- (4) A: Myosin - ATP binds to it



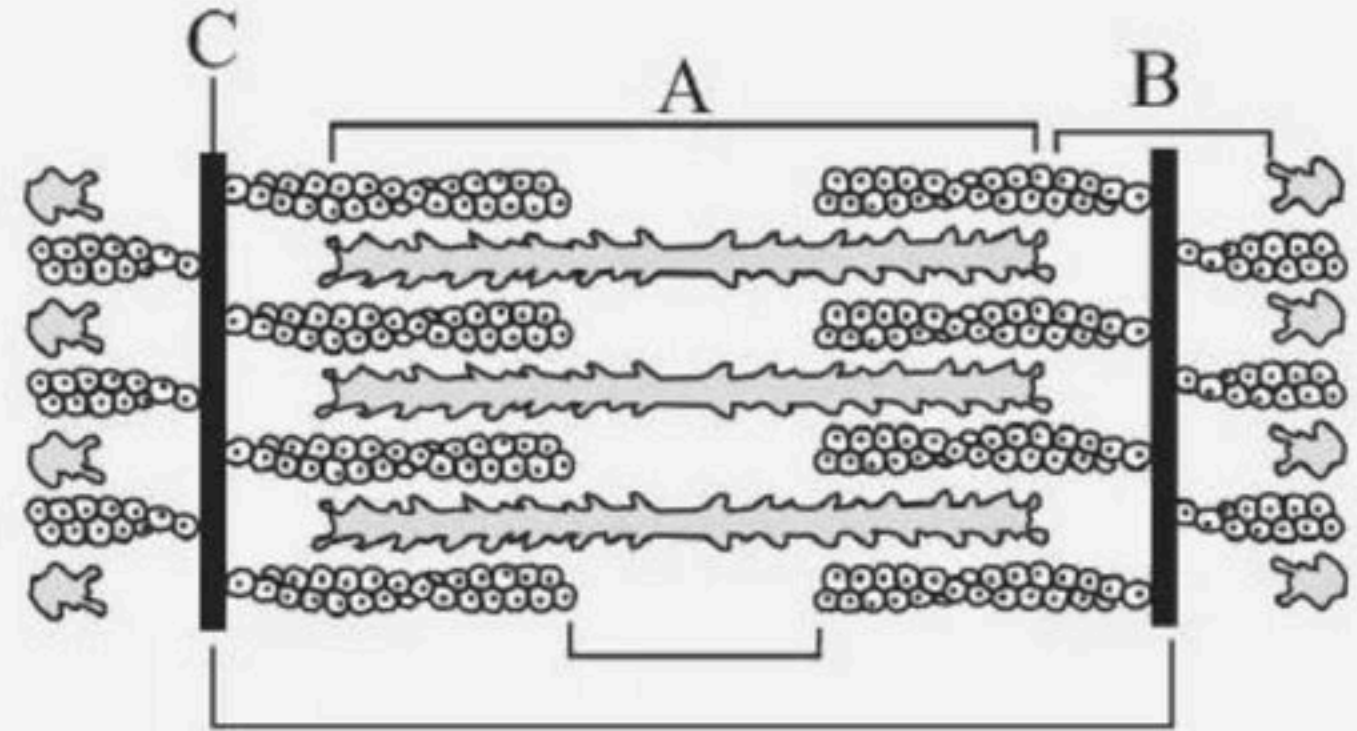
99. By referring to the given flow chart, fill up the blanks by selecting the correct option

- (1) A- Appendicular skeleton, B- skull, C- Ribs, D- Limbs
- (2) A- Thoracic skeleton, B- Limbs, C- Ribs, D- skull
- (3) Lumbar skeleton, B- Ribs, C- Limbs, D- skull
- (4) A- Thoracic skeleton, B- skull, C- Ribs, D- Limbs



100. Identify A, B and C

- (1) A - I band, B - A band, C - Z line
- (2) A - A band, B - I band, C - M line
- (3) A - A band, B - I band, C - Z line
- (4) A - I band, B - A band, C - M line



THANK YOU



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